

Mapping Wetland Inundation Regimes using Earth Observation Embeddings: A Case Study in the Florida Everglades

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Background & Motivation

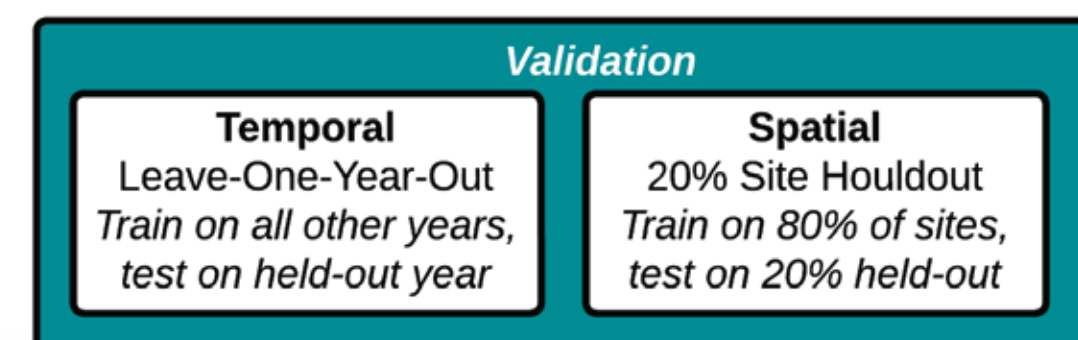
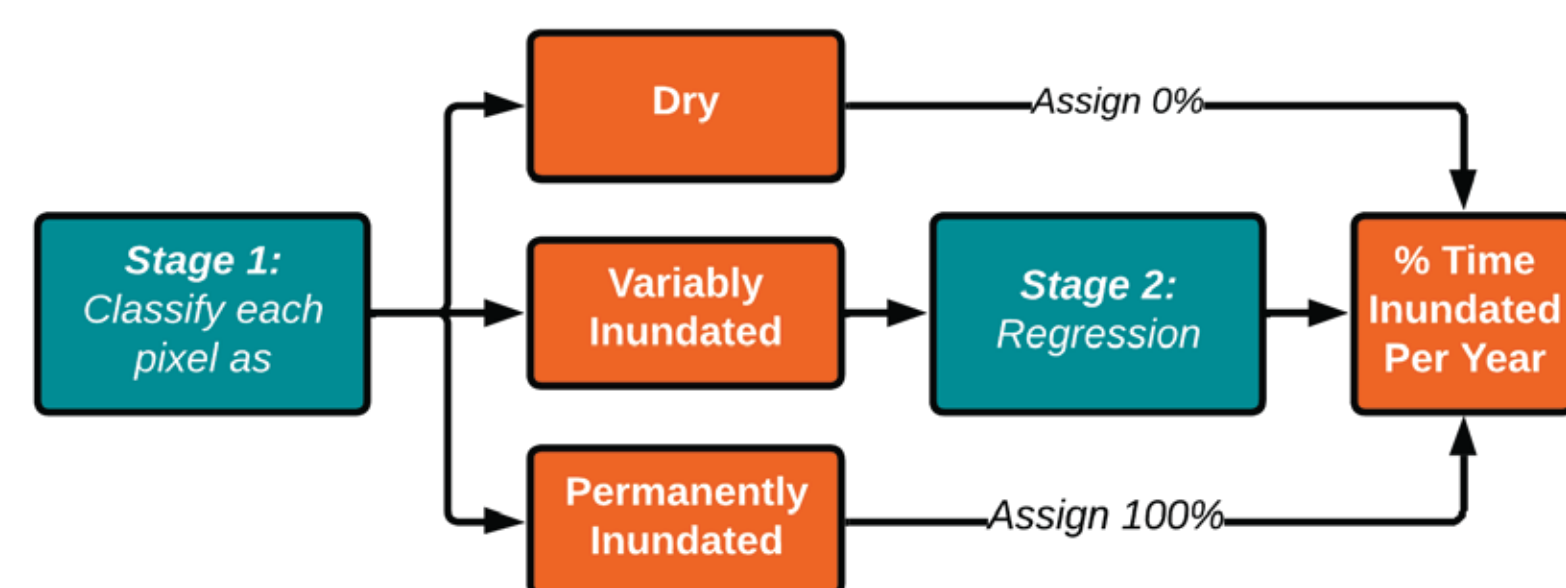
- Understanding how often wetlands are inundated is key to water resources management, biogeochemical modeling, and flood mitigation
- Wetland inundation is spectrally complex to map
- Geospatial foundation models** generate high-dimensional **embeddings**, consolidating spectral-temporal patterns into one dataset

Objective:

Evaluate if Earth observation vector embeddings outperform traditional features and baseline datasets in predicting wetland inundation regimes

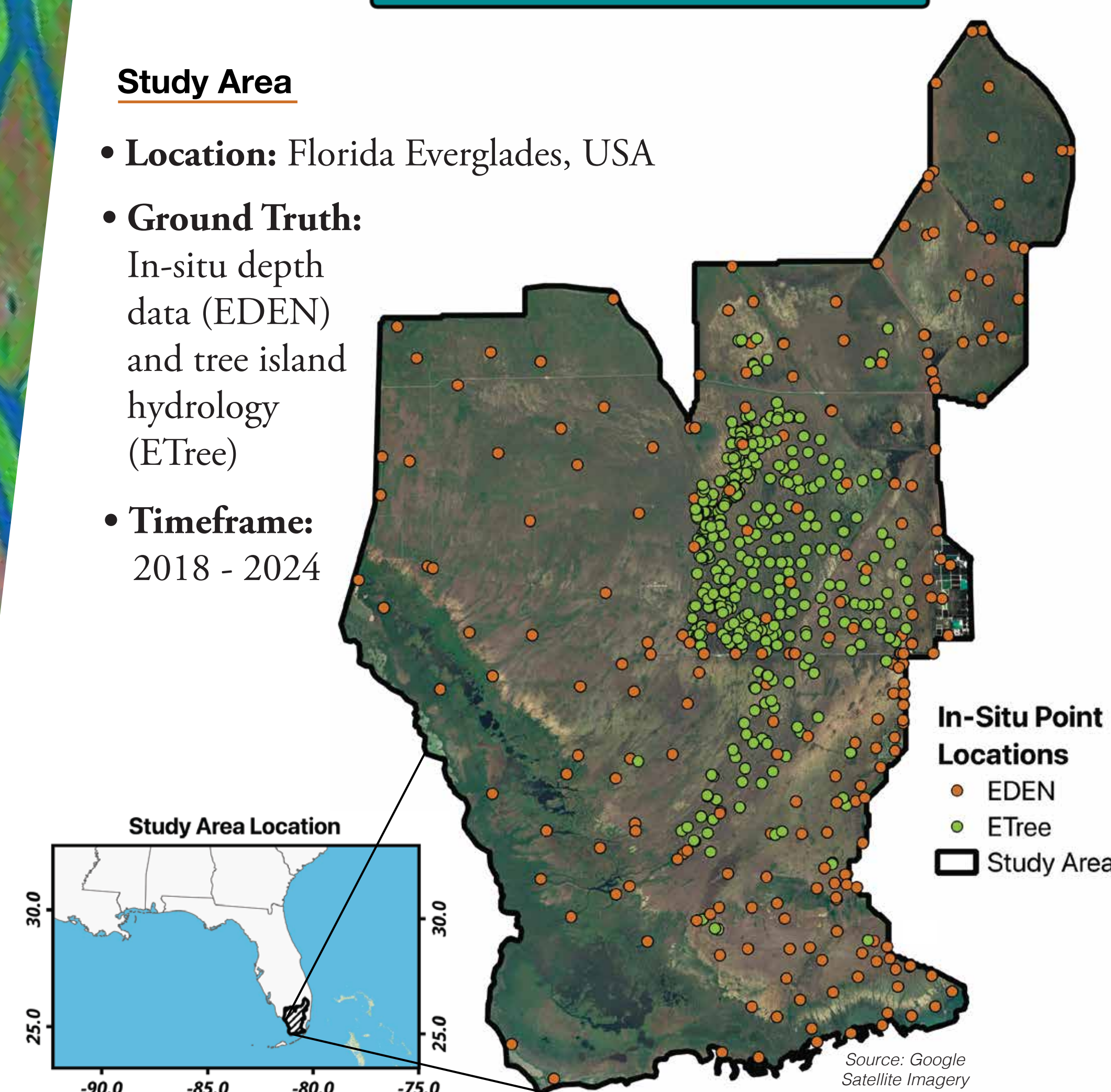
Study Area & Methods

- Classify pixels as dry, variably inundated, or permanently inundated
- Regress to predict annual % of time inundated

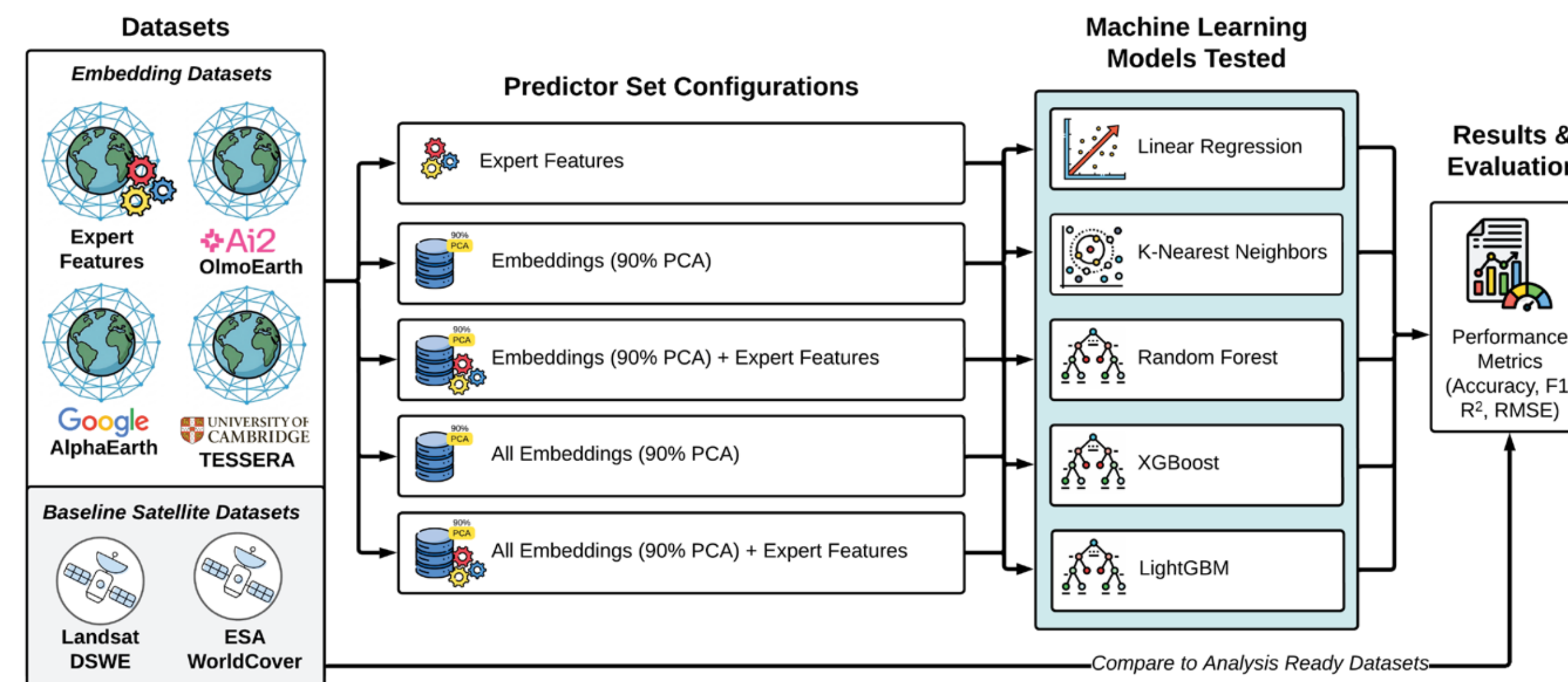


Study Area

- Location:** Florida Everglades, USA
- Ground Truth:** In-situ depth data (EDEN) and tree island hydrology (ETree)
- Timeframe:** 2018 - 2024



Technical Approach: Evaluating Embeddings



Embedding Datasets & Expert Features

Feature	AlphaEarth	OlmoEarth	TESSERA
Developer	Google DeepMind	Allen Institute for AI (AI2)	University of Cambridge
Dimensions	64	128 (Tiny) to 1,024 (Large)	128
Spatial Res.	10 m	10 m (Flexible)	10 m
Temporal Focus	Annual	Flexible	Annual
Training Datasets	Sentinel-1*, Sentinel-2*, Landsat*, ALOS PALSAR ScanSAR, Copernicus DEM, GEDI, ERA5-Land, GRACE, NLCD, Wikipedia, GBIF	Sentinel-1*, Sentinel-2*, Landsat*, WorldCover, OpenStreetMap, CDL, WorldCereal, SRTM, Canopy Height	Sentinel-1*, Sentinel-2*
Model Openness	Closed Source (Proprietary)	Open Source (Weights/Code)	Open Source (Weights/Code)
Source	Brown et al., 2025	Herzog et al., 2025	Feng et al., 2025

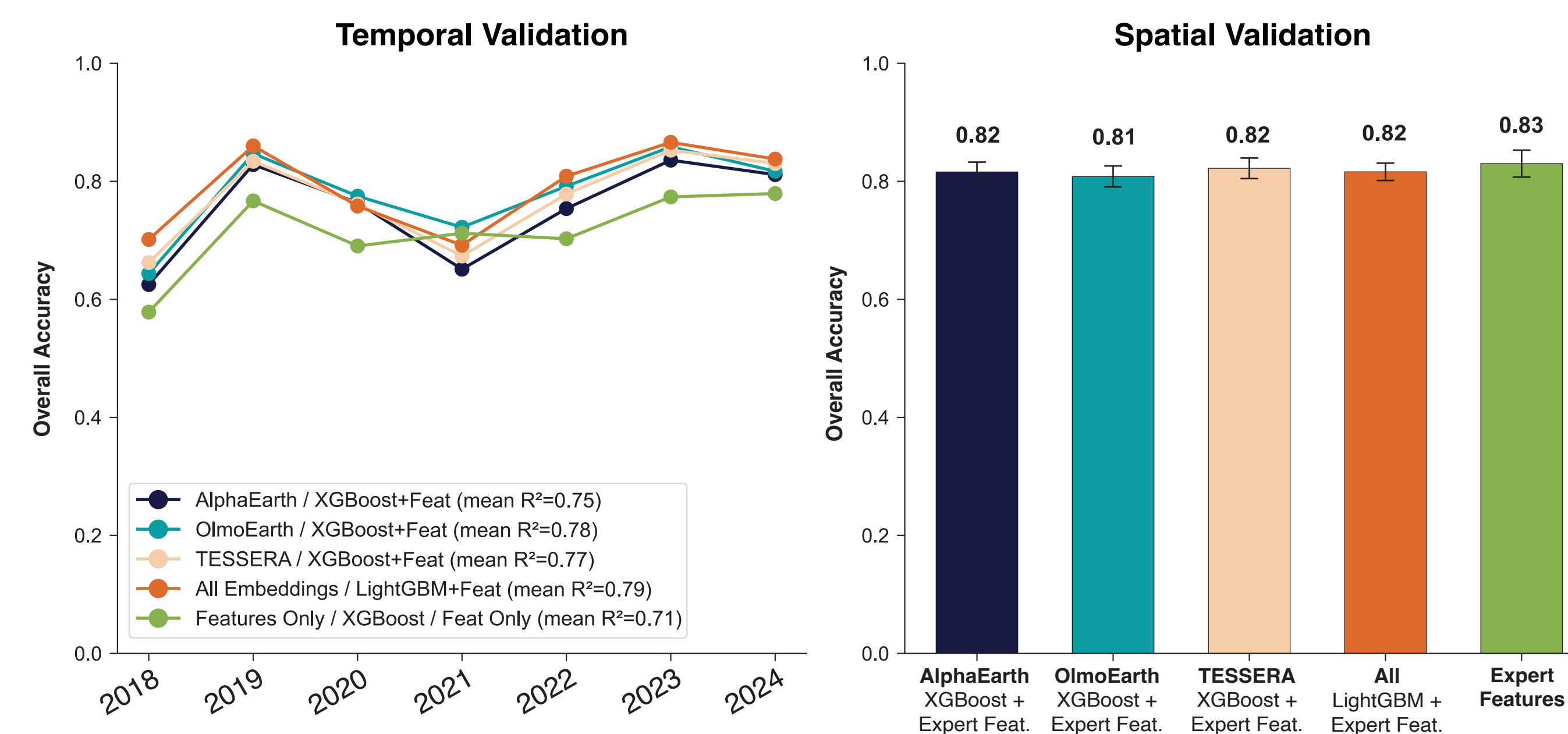
Expert Engineered Features

DAYMET (Monthly Precip, Seasonal T_{max}, & PET), Annual S2 NDWI (Median & Variance), Annual ALOS PALSAR (L -Band), Topographic Wetness Index (Derived from 10 m DEM), in-situ reported elevation, SSURGO (bulk density, K_{sat}, soil name)

* indicates this data source is required for inference

Stage 1: Classification Results

Models with Embeddings Can Classify Inundation Regimes with Mean Overall Accuracy ≥ 0.75



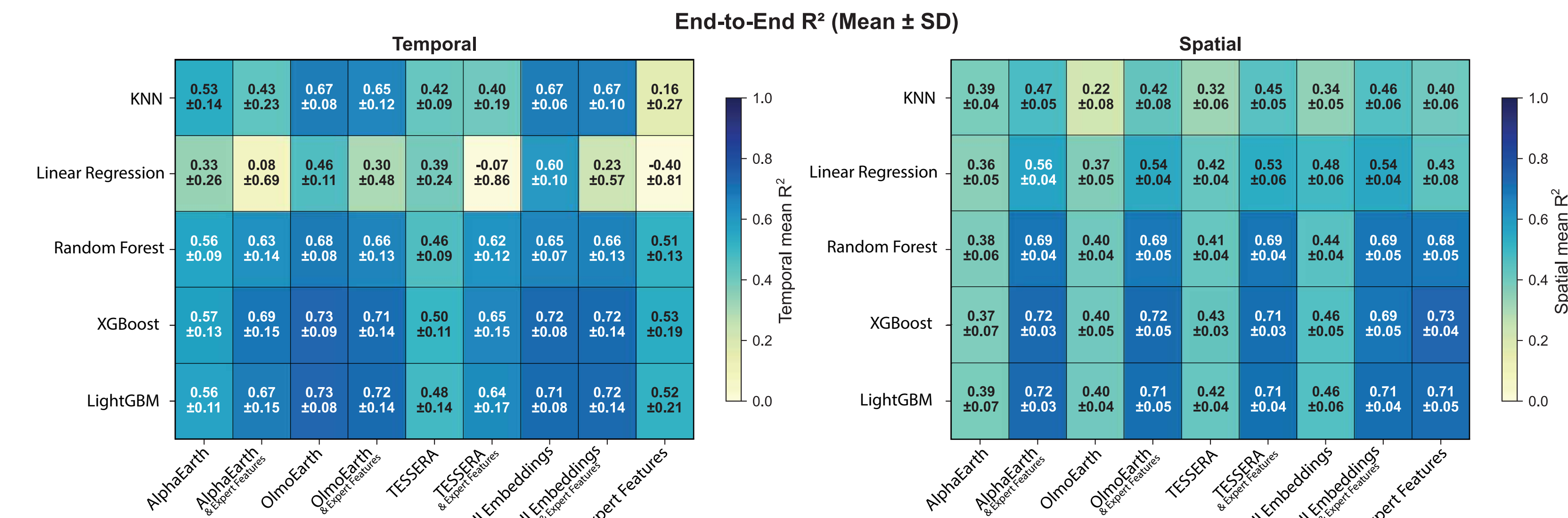
Both Embeddings and Expert Features Outperform Baseline Satellite Datasets

Baseline	Accuracy		Δ AlphaEarth		Δ OlmoEarth		Δ TESSERA		Δ Feat.	
	T	S	T	S	T	S	T	S	T	S
DSWE [1]	0.28	0.28	+0.43	+0.42	+0.52	+0.40	+0.47	+0.47	+0.43	+0.55
DSWE [1-2]	0.29	0.29	+0.42	+0.41	+0.51	+0.39	+0.46	+0.46	+0.42	+0.54
DSWE [1-3]	0.40	0.41	+0.31	+0.29	+0.40	+0.27	+0.35	+0.34	+0.31	+0.42
DSWE [1-4]	0.64	0.64	+0.07	+0.06	+0.16	+0.04	+0.11	+0.11	+0.07	+0.19
Dynamic World	0.52	0.52	+0.19	+0.18	+0.28	+0.16	+0.23	+0.23	+0.19	+0.31

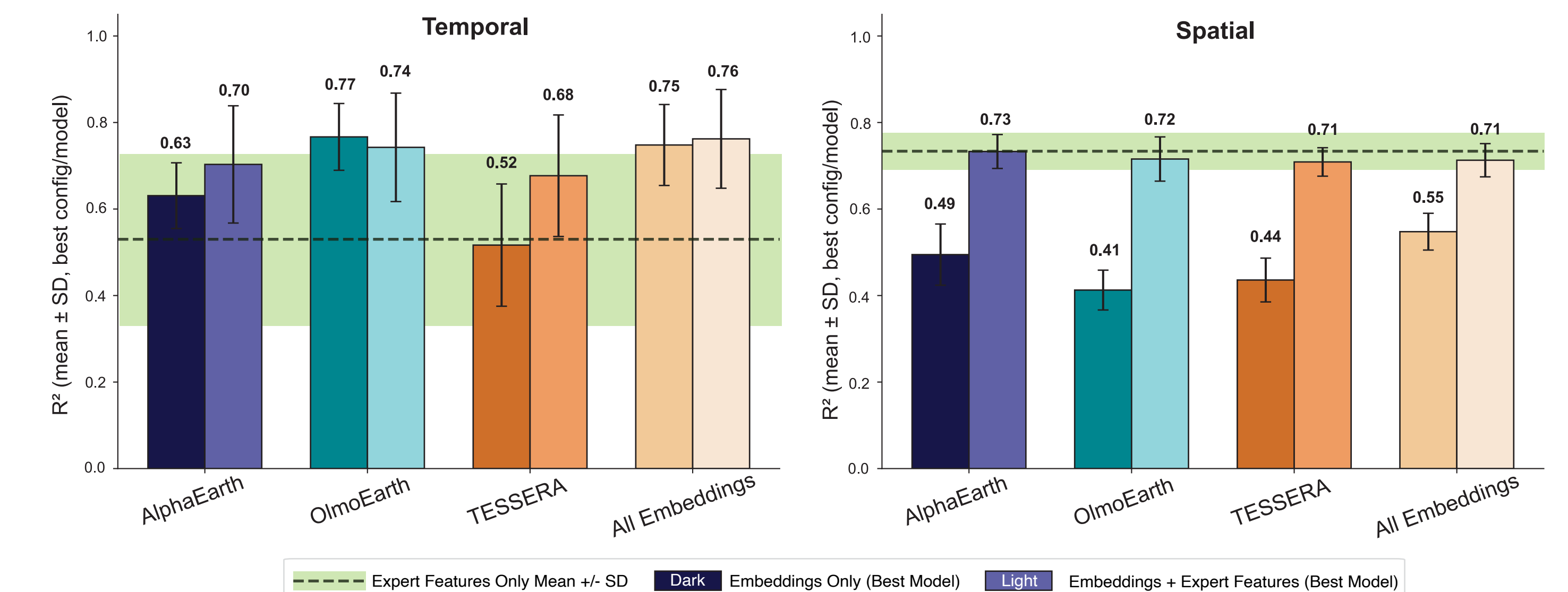
T = Temporal, S = Spatial; Δ columns represent increase in accuracy compared to baseline

Stage 2: Regression Results

Embeddings Combined with Expert Features Can Predict % Inundated with $R^2 > 0.7$, Tree-Based Models Perform Best

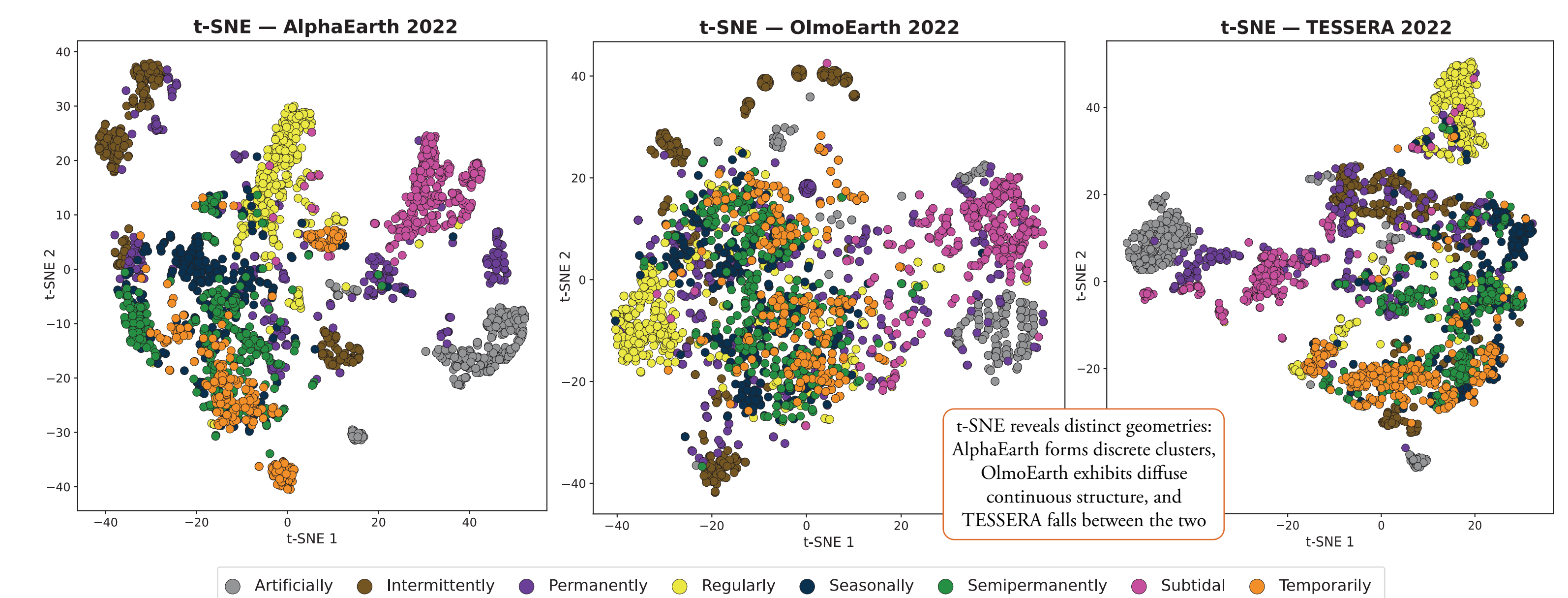


Embeddings Increase Predictive Performance Across Different Years, but Expert Features Drive Spatial Performance



Visualizing Inundation Regimes in Embeddings

- Results of extracting 300 points of each National Wetland Inventory water regimes across the study area



Key Takeaways

- Embeddings perform best when combined with additional hydrological features, with tree-based models achieving the highest metrics
- For temporal validation, embeddings are more effective than expert features at detecting annual-scale variations in inundation regimes.
- For spatial validation, expert engineered features proved sufficient for local/regional scales like our Everglades site but further testing is required to determine which is best for multi-site generalization

